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The anaerobic power reserve and its applicability in professional road cycling

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ABSTRACT

This study examined if short-duration record power outputs can be predicted with the Anaerobic Power Reserve (APR) model in professional cyclists using a field-based approach. Additionally, we evaluated if modified model parameters could improve predictive ability of the model. Twelve professional cyclists ($\text{VO}_{2\text{max}} 75 \pm 6 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) participated in this investigation. Using the mean power output during the last stage of an incremental field test, sprint peak power output and an exponential constant describing the decrement in power output over time, a power-duration relationship was established for each participant. Record power outputs of different durations (5 to 180 s) were collected from training and competition data and compared to the predicted power output from the APR model. The originally proposed exponent ($k = 0.026$) predicted performance within an average of 43 W (Standard Error of Estimate (SEE) of 32 W) and 5.9%. Modified model parameters slightly improved predictive ability to a mean of 34–39 W (SEE of 29 – 35 W) and 4.1 – 5.3%. This study shows that a single exponent model generally fits well with the decrement in power output over time in professional cyclists. Modified model parameters may contribute to improving predictability of the model.

ARTICLE HISTORY

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KEYWORDS

Performance; endurance; cycling; power output

Introduction

Competitive road cycling is predominantly an endurance-based aerobic sport (Lucia, Hoyos, & Chicharro, 2001). However, it's conceivable that decisive moments within a competition also require the cyclists to possess a high anaerobic capacity. The ability to generate high power outputs over a short period of time is a decisive performance parameter in cycling to close a gap, break away from the pack or win a sprint (Abbiss, Menaspas, Villerius, & Martin, 2013; Rønnestad & Mujika, 2013). High-intensity interval training (HIT) has been shown to improve both indices of aerobic and anaerobic capacity in well-trained cyclists (Laursen, Shing, Peake, Coombes, & Jenkins, 2005; Lindsay et al., 1996). Therefore, besides low-intensity aerobic training, HIT is also an important part of cyclists' training programme to prepare optimally for key moments in the race and to improve the physiological potential of the athlete (Laursen & Jenkins, 2002). There are several measures proposed in the literature with which to prescribe exercise intensity for intervals during HIT such as a power output, heart rate or rating of perceived exertion (Buchheit & Laursen, 2013b). Given the well-known lag in heart rate response at the onset of exercise, and the decreased heart rate recovery response during accumulated intervals (Buchheit & Laursen, 2013b), heart rate may be less applicable as a measure to monitor and/or prescribe intensity during HIT, especially for short duration intervals (< 3 min). In contrast, power output provides a direct and objective measure of exercise intensity (Sanders, Myers, & Akubat, 2017). Ranges in power output for supramaximal HIT intervals (i.e. training at intensities > power output at $\text{VO}_{2\text{max}}$) are typically

estimated based on functional threshold power (FTP) or power output at $\text{VO}_{2\text{max}}$ (Allen & Coggan, 2010; Billat, 2001; Laursen & Jenkins, 2002). However, two cyclists with a similar power output at $\text{VO}_{2\text{max}}$ can have significantly different sprint peak power outputs (Sanders, Heijboer, Akubat, Meijer, & Hesselink, 2017; Weyand, Lin, & Bundle, 2006). As such, if the intensity of a HIT session is exclusively based on an "aerobic marker" (e.g. intervals at 130% of power output at $\text{VO}_{2\text{max}}$) the athlete with the higher sprint peak power output has to work at a lower percentage of their maximal capacity, potentially resulting in a different physiological demand and exercise tolerance (Buchheit & Laursen, 2013b).

The anaerobic reserve could be useful in individualising exercise intensity for HIT. In cycling, the anaerobic power reserve (APR) is defined as the difference between maximal sprint peak power output and power output at $\text{VO}_{2\text{max}}$ (Weyand et al., 2006). Studies have used the anaerobic reserve range to set out the minimal and maximal values of a short-duration power-duration curve (Bundle, Hoyt, & Weyand, 2003; Sanders et al., 2017; Weyand & Bundle, 2005; Weyand et al., 2006). Subsequently, an exponential decay model is used to describe the decrement in power output over time. The obtained power-duration curve can be used to predict power output over all-outs efforts lasting from a few seconds to a few minutes (Sanders et al., 2017; Weyand et al., 2006). This is a similar approach to the critical power (CP) model (Poole, Burnley, Vanhatalo, Rossiter, & Jones, 2016), however, the CP model mainly applies to longer duration performances (approximately 3 – 45 min) while the APR model focuses on short-duration performance (5 – 300 s). The APR model is based on the assumption that the decrement in (cycling)

power output over time in humans, irrespective of between-individual differences in absolute power outputs, is the same when this is expressed in relation to their anaerobic reserve (Bundle et al., 2003; Weyand et al., 2006).

Bundle and colleagues (Bundle et al., 2003; Bundle & Weyand, 2012; Weyand & Bundle, 2005; Weyand et al., 2006) used the anaerobic reserve model to predict performance during all-out efforts of different durations in runners and cyclists. To date, most studies have used recreationally active runners (Bundle et al., 2003; Bundle & Weyand, 2012; Weyand & Bundle, 2005) or cyclists (Weyand et al., 2006) in a laboratory setting. However, in elite environments, field-testing is more representative of the athlete's environment and often more valued by coaches and practitioners as field tests are easier to implement in to daily practice. Recently, in a preliminary pilot study, we presented four case studies on the applicability of the APR model in professional cyclists using a field-based approach (Sanders et al., 2017). It was shown that the power output predicted by the model was very largely to nearly perfectly correlated to the actual power output obtained during all-out time trials for each cyclist ($r = 0.88 - 0.97$). Even though these results should be considered promising, it still remains questionable how well the model would fit a larger group of (elite) athletes. However, the implementation of multiple all-out time trials to a professional cyclists' training plan would be too intrusive and would limit the possibility to collect data with a larger pool of athletes. Therefore, the record power outputs (Allen & Coggan, 2010; Pinot & Grappe, 2011) over short-durations (5 – 180 s) achieved during training and competitions can be assessed and would provide the possibility to collect a substantial dataset that provides an indication of short-duration performance of the cyclists.

Therefore, the aim of this study was to test the ability of the APR model in predicting record power outputs achieved during training and racing in professional cyclists using a field-based approach. In addition, this study aims to evaluate if modifications to the previously established exponential constant ($k = 0.026$) (Weyand et al., 2006) or modifications in the parameters used to determine the anaerobic power reserve could improve predictive ability of the APR model in this cohort of professional cyclists.

Methods

Participants

Twelve professional cyclists from a Union Cycliste Internationale (UCI) World-Tour professional cycling team participated in this investigation (mean $\pm$ SD: age 29 ± 5 y, height 1.81 ± 0.06 m, body mass 72.3 ± 5.3 kg, $\dot{V}O_{2\max}$ 74.6 ± 5.9 ml·kg⁻¹·min⁻¹). Participants were informed of the purpose and procedures of the investigation. Institutional ethics approval was granted and in agreement with the Helsinki Declaration.

Testing

Every cyclist performed an incremental field test protocol (Sanders et al., 2017) during a January training camp. The protocol consisted of 6 times 6 min blocks on uphill terrain

(mean gradient of 4.8%). The cyclists were advised to maintain a set power output during the 6 min intervals with the defined power output increasing with every interval (mean increment of 23 ± 9 W per interval). The absolute power output and stage increments were based on pre-season laboratory testing (data not shown). After the 6 min interval the riders had 6–10 min active recovery ($< 55\%$ of power output at $4 \text{ mmol}\cdot\text{L}^{-1}$) before starting the next interval. The last effort was a 6 min all-out performance; mean power output during this 6 min bout was used as the lower bound of the anaerobic power reserve (PO_{incr}). Previous research has typically used power output at $\dot{V}O_{2\max}$ as the lower bound of the anaerobic reserve, however since we used a field-based protocol, we adopted a different approach. Nevertheless, studies have shown that all-out performances of ~ 5 min largely correlates to power at $\dot{V}O_{2\max}$ as time to exhaustion at $\dot{V}O_{2\max}$ varies between 4–8 min (Berthon et al., 1997; Hill & Rowell, 1996). Power output and cadence were measured (1 Hz) using a mobile ergometer system (Pioneer Power Meter, Kawasaki, Kanagawa, Japan). Riders were instructed to perform zero-offset procedures prior to each training session or stage according to manufacturers' instructions. Sprint power output (PO_{sp}) was defined as the peak power output the cyclists could achieve during all-out sprints in the field during training sessions, measured as a 1-second peak power output. The sprints were performed as 10 s "flying sprints" with the cyclist already riding at $30\text{--}35 \text{ km}\cdot\text{h}^{-1}$ and the sprints were performed with a self-selected cadence.

Original anaerobic power reserve approach

With PO_{sp} and PO_{incr} as the maximal and minimum values of the curve, a power-duration relationship was established for each subject individually (Bundle & Weyand, 2012). This relationship was set using the following formula (Equation (1); Bundle et al., 2003):

$$PO_t = PO_{\text{incr}} + (PO_{\text{sp}} - PO_{\text{incr}}) * e^{(-k*t)} \quad (1)$$

Where t is the duration of the all-out trial, PO_t is the power output maintained for that trial with a duration of t , PO_{incr} is the mean power output during the last stage of the incremental test, PO_{sp} is the sprint peak power output, e is the base of the natural logarithm and k is the exponent that describes the decrement in power output over time. The exponential time constant ($k = 0.026$) is based on the previously established exponential power-duration curve fitted through data in recreationally active cyclists (Weyand et al., 2006) and tested in professional cyclists (Sanders et al., 2017).

Modified approaches

In addition to the original modelling approach described above, four modified approaches were adopted to evaluate if predictive ability could be improved with modified model parameters. In "Modified Approach 1", the lowest bound of the anaerobic reserve (PO_{incr}) was replaced with record power output over 3 min ($PO_{3\text{min}}$) achieved during the study period (Equation (2)).

$$PO_t = PO_{3min} + (PO_{sp} - PO_{3min}) * e^{(-k_2 * t)} \quad (2)$$

In “Modified Approach 2”, the original exponential decay constant ($k_1 = 0.026$) was replaced with a modified exponent (k_2). In order to evaluate if the previously established exponential time decay constant could be optimised, an iterative best-fit approach was adopted for every participant using their measured record power outputs, assessed values for PO_{incr} and PO_{sp} and Equation (1). The iterative best-fit approach was performed using a least-squares analysis in which an optimal time decay constant could be determined for every individual by aiming to minimise the residual sum of squares (RSS; sum of the squared differences) between predicted power output by the model and record power outputs. Based on the optimal individual exponential decay constants, a general time decay constant was established by taking the mean of the individual constants, a similar approach to previous studies (Weyand & Bundle, 2005; Weyand et al., 2006). The modified constant calculated using this approach was $k_2 = 0.0244 \pm 0.0053$. In addition, this procedure was repeated for Equation (2) with PO_{3min} as the lowest anchor point in the iterative process instead of PO_{incr} . The modified constant (k_3) calculated using this approach was $k_3 = 0.0277 \pm 0.0061$. In “Modified Approach 3”, both the modified lowest bound of the anaerobic reserve (PO_{3min}) as well as exponential decay constant k_2 were adopted. Lastly, “Modified Approach 4” used PO_{3min} as the lower bound of the anaerobic reserve and exponential decay constant k_3 .

Record power outputs

In the four months following the incremental exercise test protocol, record power outputs of different durations achieved during training and competitions were obtained for each cyclist, using previously described methods (Allen & Coggan, 2010; Pinot & Grappe, 2011). The chosen durations for the record power outputs were: 5, 10, 15, 30, 45, 60, 90, 120, 150 and 180 seconds, respectively. Measured record power outputs were compared with the power output predicted by the APR modelling approaches.

Statistical analysis

Record power outputs achieved over the different durations during training and racing was compared to the predicted power outputs by the original approach and Modified Approach 1,2,3 and 4 using a multilevel random intercept model with Tukey’s method for pairwise comparisons, using the statistical package R (R: A Language and environment for statistical computing, Vienna, Austria). To account for individual differences in absolute power outputs, random effect variability was modelled using a random intercept for each individual participant. Level of significance was established at $P < 0.05$. Standardised effect size is reported as Cohen’s d , using the pooled standard deviation as the denominator. Qualitative interpretation of d was based on the guidelines provided by Hopkins, Marshall, Batterham, and Hanin (2009): 0 – 0.19 trivial; 0.20 – 0.59 small; 0.6 – 1.19 moderate; 1.20 – 1.99 large; ≥ 2.00 very large. Standard Error of Estimate (SEE) was

calculated to evaluate the accuracy of the predictions between modelled and record power output provided by the regression. Bland-Altman plots including mean bias and 95% limits of agreement were used to visually represent the differences between record power outputs and predicted power outputs. The limits of agreement were calculated according to the recommendations by Bland and Altman (2007).

Results

Absolute and relative mean power output during the last stage of the incremental test (PO_{incr}), PO_{3min} and sprint peak power output (PO_{sp}) as well as the anaerobic reserve can be found in Table 1. Mean and maximal cadence during the 10 s sprints was 103 ± 10 rev·min⁻¹ and 113 ± 12 rev·min⁻¹. PO_{sp} were achieved at a cadence of 103 ± 9 rev·min⁻¹.

A total of 1647 training and race files were analysed for the 12 participants, averaging 137 ± 22 sessions per participant. Record power outputs varying from 5 up to 180 seconds were collected for every individual over the course of the study period. Mean cadence during the 5 and 10 s record power outputs (102 ± 7 and 101 ± 7 rev·min⁻¹, respectively) was *moderately* to *largely* higher ($d = 0.95 - 1.88$) compared to cadences during the 30 – 180 s record power outputs (mean cadence ranging between $87 - 93$ rev·min⁻¹). *Trivial* to *small* differences were observed for most cadences achieved between 30 – 180 s record power outputs ($d = 0.01 - 0.58$) with only the cadence for 45 s record power output (93 ± 9 rev·min⁻¹) being *moderately* higher ($d = 0.80$) compared to cadence at the 150 s record power output (85 ± 11 rev·min⁻¹).

The predictive ability of the APR model using the original approach and the four modified approaches proposed in this study are presented in Table 2. Using the previously established exponential decay constant ($k = 0.026$), record power outputs remained within an average of 5.9% and an average of 43 W of the predicted power output by the model (SEE 32 ± 19 W, $R^2 = 0.97$). Figure 1 presents the record power outputs compared to the predicted line of identity (i.e. predicted power output = actual power output) by the APR model using the original approach as well as the respective Bland-Altman plot. Predictive ability of the APR model was slightly improved using the modified approaches with mean deviation between predicted power output and record power output being lower in the modified approaches ($d = 0.22 - 0.56$, ES = small). Figure 2 presents the record power outputs

Table 1. The field-derived mean power output during the last stage of the incremental test, record power output over 3 min and sprint peak power output achieved by the participants.

n = 12	Mean $\pm$ SD	Range
Mean PO during the last stage of incremental test (PO_{incr}) (W)	458 $\pm$ 29	401 – 505
Mean PO during the last stage of incremental test (PO_{incr}) (W·kg ⁻¹)	6.35 $\pm$ 0.39	5.69 – 7.05
Record PO over 3min (PO_{3min}) (W)	500 $\pm$ 43	440 – 563
Record PO over 3min (PO_{3min}) (W·kg ⁻¹)	6.92 $\pm$ 0.26	6.56 – 7.36
Sprint peak PO (PO_{sp}) (W)	1254 $\pm$ 153	1064 – 1635
Sprint peak PO (PO_{sp}) (W·kg ⁻¹)	17.37 $\pm$ 1.84	15.80 – 22.71
APR for PO_{incr} (W)	796 $\pm$ 141	622 – 1150
APR for PO_{3min} (W)	753 $\pm$ 111	602 – 1141

Abbreviations: PO, power output; APR, anaerobic power reserve

Table 2. Predictive ability of the Anaerobic Power Reserve model with the original and modified approaches.

Participant	Original model PO_{incr} PO_{sp} $k_1 = 0.026$		Modified Approach 1 $\rightarrow PO_{3min}$ PO_{sp} $k_1 = 0.026$		Modified Approach 2 PO_{incr} PO_{sp} $\rightarrow k_2 = 0.0244$		Modified Approach 3 $\rightarrow PO_{3min}$ PO_{sp} $\rightarrow k_2 = 0.0244$		Modified Approach 4 $\rightarrow PO_{3min}$ PO_{sp} $\rightarrow k_3 = 0.0277$	
	Mean deviation (W)	SEE (W)	Mean deviation (W)	SEE (W)	Mean deviation (W)	SEE (W)	Mean deviation (W)	SEE (W)	Mean deviation (W)	SEE (W)
1	48	16	25	15	38	16	22	16	44	16
2	49	32	29	29	42	36	30	32	30	27
3	67	24	30	22	56	20	21	18	39	27
4	62	23	28	19	54	28	40	24	20	17
5	23	22	27	21	29	26	35	26	20	17
6	17	17	14	16	16	18	22	17	10	17
7	64	76	80	71	72	82	85	69	71	66
8	33	52	33	52	33	57	36	57	32	48
9	46	51	47	51	39	50	36	49	45	52
10	42	14	31	13	33	18	33	18	38	12
11	30	36	32	34	30	41	41	40	25	29
12	33	19	31	18	30	23	33	22	33	16
Mean $\pm$ SD	43 $\pm$ 16	32 $\pm$ 19	34 $\pm$ 16	30 $\pm$ 18	39 $\pm$ 15	35 $\pm$ 20	36 $\pm$ 17	32 $\pm$ 18	34 $\pm$ 16	29 $\pm$ 17
Mean deviation	5.9% $\pm$ 2.5%		4.3% $\pm$ 2.0%		5.3% $\pm$ 2.2%		4.6% $\pm$ 2.1%		4.1% $\pm$ 1.9%	

Abbreviations: PO_{incr} , mean power output over the last 6 min stage of the incremental field tests; PO_{3min} , record power output over 3 min; SEE, standard error of estimate; SD, standard deviation; k_1 , exponential decay constant proposed by Weyand et al. (2006); k_2 , modified exponential decay constant determined with the individual best fits to the record power outputs using Equation (1); k_3 , modified exponential decay constant determined with the individual best fits to the record power outputs using Equation (2)

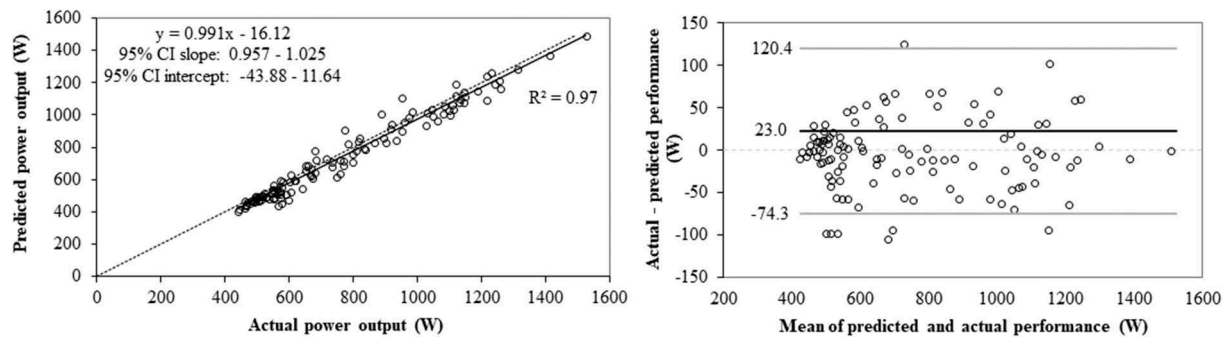


Figure 1. Record power output versus those predicted by the APR model using the original approach (a) and Bland-Altman plot (b), displaying bias (black line) and 95% limits of agreement (grey lines), comparing record power output for the different durations versus those predicted by the APR model using the original approach.

compared to the predicted line of identity (i.e. predicted power output = actual power output) ($R^2 = 0.96 - 0.97$) by the APR model using the four modified approaches (A – D) as well as their respective Bland-Altman plots (E – H).

Record power outputs achieved during the training and races and predicted power outputs by different modelling approaches are presented in Table 3 and the respective plots of predicted and actual power output for a participant are presented in Figure 3. The multilevel model analysis revealed no significant differences ($P > 0.75$) between record power outputs and predicted power output by the modelling approaches, for all durations. *Trivial* to *small* differences ($d = 0.03 - 0.56$) were observed when comparing predicted power output by the original approach to record power output over durations varying from 5 to 60 s, whilst these differences were *moderate* ($d = 0.59 - 1.01$) for power outputs over durations from 90 to 120 s. *Trivial* to *small* differences ($d = 0.08 - 0.30$) were observed when comparing record power output to predicted power output by Modified Approach 1, over all durations. For Modified Approach 2, *trivial* to *small* differences ($d = 0.09 - 0.29$) were observed for durations from 5 to 90 s, whilst the difference were *moderate* ($d = 0.75 - 0.93$) for durations 120 to 180 s. For Modified Approach 3, a *moderate* difference ($d = 0.65$) was observed between record power output and predicted power output over a duration of 45 s, whilst all the other durations showed *trivial* to *small* differences ($d = 0.04 - 0.52$). For Modified Approach 4, *trivial* to *small* differences ($d = 0.03 - 0.30$) were observed comparing record power output to predicted power output, over all durations.

Discussion

This study aimed to test the ability of the APR model in professional cyclists in predicting record power outputs achieved during training and competitions over different durations (5 – 180 s). In addition, it was examined if modified model parameters could improve predictive ability of the APR model. Firstly, it was shown that in professional cyclists, short-duration performance achieved in training and competitions can be predicted from field-based measurements of the maximal sprint peak power and mean power output during the last stage of an incremental test,

using the original APR model ($k = 0.026$). The single exponential model generally fits well with the decrement in power versus duration for this elite athlete population. In addition, it was shown that adjusting model parameters, such as a modified exponential constants or modified lowest bound of the anaerobic reserve (i.e. PO_{3min}), led to slight improvements in predictive ability.

Using the originally proposed exponential constant ($k = 0.026$) the APR model predicted the short-duration performances within an average of 43 W and 5.9%. The predictive ability of the model in this study is found to be slightly better than our previously reported predictive ability of 6.6% and 53 W in a smaller cohort of professional cyclists (Sanders et al., 2017). Furthermore, its predictive ability is lower compared to the previously determined predictive ability of the model in running (3.7% and 2.6%) (Bundle et al., 2003; Weyand & Bundle, 2005) and slightly higher compared to predictability previously reported for cycling performance (6.6%) (Weyand et al., 2006). The described accuracy of the model is within an average of 43 W whilst Weyand et al. (2006) showed a slightly lower mean deviation of 34 W. In addition, the proposed modified exponential constant ($k_2 = 0.0244$), based on an iterative best-fit procedure, improved overall predictive ability of the model slightly. The modified constant predicted performance within a mean of 34 W and 4.3% suggesting that the methods used to obtain a modified exponential constant can improve the overall predictability of the APR model. Interestingly, it was shown that using different lower bounds of the anaerobic reserve (i.e. PO_{3min} vs PO_{3min}) resulted in different “optimal” exponents ($k_3 = 0.0277$ vs $k_2 = 0.0244$) when applying the iterative best-fit procedures, with one being higher and one being lower than the originally proposed exponent ($k_1 = 0.026$). This suggests that the exponent may vary depending on what measure as the lowest bound of the anaerobic reserve is used. When applying Modified Approach 4, which uses both a modified lowest bound of the anaerobic reserve (i.e. PO_{3min}) and the modified exponent ($k_3 = 0.0277$) based on Equation (2), predictive ability was also improved (34 W, 4.1%) compared to the original model (43 W, 5.9%).

Modifying the lowest bound of the anaerobic reserve from PO_{incr} to PO_{3min} had the biggest impact on improving the predictive ability of the model. Expectedly, this occurred

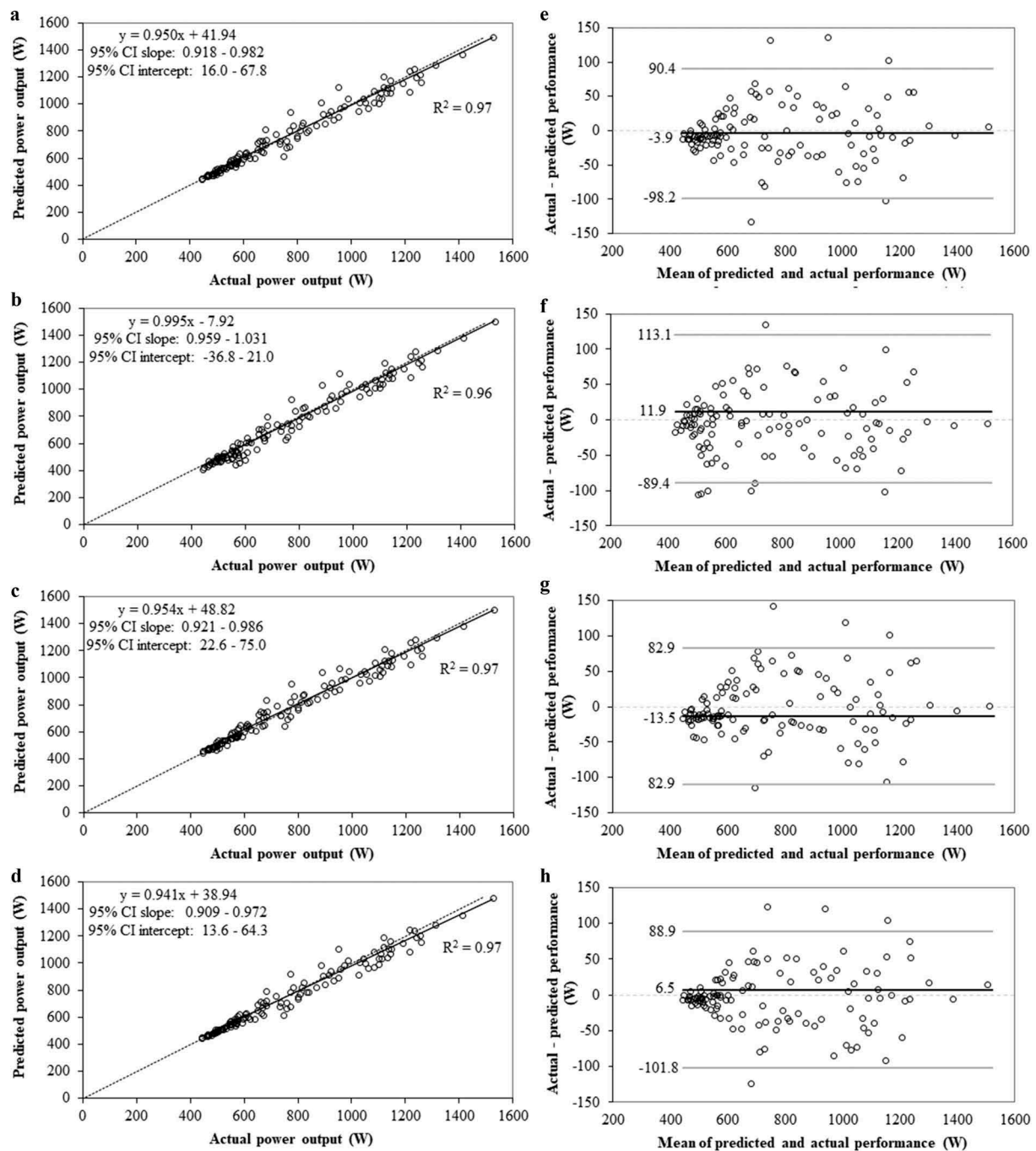


Figure 2. Record power output versus those predicted by the APR model using modified approach 1 (a), modified approach 2 (b), modified approach 3 (c) and modified approach 4 (d) and Bland-Altman plots, displaying bias (black line) and 95% limits of agreement (grey lines), comparing record power output for the different durations versus those predicted by modified approach 1 (e), modified approach 2 (f), modified approach 3 (g) and modified approach 4 (h).

at the lower intensity end of the power duration curve (90 – 180 s). Whilst *moderate* differences were observed between record power output achieved over durations of 90 to 180 s and power output predicted by the original model ($d = 0.59 - 1.01$) these differences were *trivial* to *small* when PO_{3min} was used as the lowest anchor point ($d = 0.13 - 0.29$) (“Modified Approach 1”). These results may suggest that using PO_{3min} as the lowest bound of the anaerobic reserve in the APR model could be preferred in professional cyclists. This further suggests that the applicability of the APR model in predicting power outputs over durations longer than

180 s is most likely minimal. Therefore, the APR model should be considered as a potential tool for predicting and monitoring performance (progression) within the investigated durations (5 – 180 s). When aiming to predict performance of durations longer than 3 min, the CP model might be considered favourable.

Substantial between-individual differences in mean deviation between predicted power output by the APR model and the record power outputs achieved in competition were observed (Table 2). It must be recognised that using a general exponential value for the decrement in power output over

Table 3. Record and predicted power outputs over different durations as well as the predicted power outputs by the original model and four modified approaches.

		Original model PO_{incr} PO_{sp} $k_1 = 0.026$	Modified Approach 1 $\rightarrow PO_{3min}$ PO_{sp} $k_1 = 0.026$	Modified Approach 2 PO_{incr} PO_{sp} $\rightarrow k_2 = 0.0244$	Modified Approach 3 $\rightarrow PO_{3min}$ PO_{sp} $\rightarrow k_2 = 0.0244$	Modified Approach 4 $\rightarrow PO_{3min}$ PO_{sp} $\rightarrow k_3 = 0.0277$
	Record power output (W)	Predicted power output (W)	Predicted power output (W)	Predicted power output (W)	Predicted power output (W)	Predicted power output (W)
5 s	1210 $\pm$ 134	1173 $\pm$ 132	1177 $\pm$ 133	1179 $\pm$ 134	1183 $\pm$ 134	1170 $\pm$ 135
10 s	1110 $\pm$ 116	1086 $\pm$ 118	1095 $\pm$ 120	1096 $\pm$ 120	1105 $\pm$ 121	1083 $\pm$ 120
15 s	1013 $\pm$ 107	1009 $\pm$ 105	1023 $\pm$ 107	1023 $\pm$ 107	1030 $\pm$ 105	1008 $\pm$ 106
30 s	831 $\pm$ 69	831 $\pm$ 75	853 $\pm$ 80	849 $\pm$ 78	871 $\pm$ 82	835 $\pm$ 78
45 s	714 $\pm$ 66	711 $\pm$ 56	739 $\pm$ 63	729 $\pm$ 59	756 $\pm$ 66	722 $\pm$ 61
60 s	661 $\pm$ 69	629 $\pm$ 44	662 $\pm$ 54	646 $\pm$ 45	677 $\pm$ 55	646 $\pm$ 52
90 s	560 $\pm$ 47	536 $\pm$ 33	574 $\pm$ 46	548 $\pm$ 35	584 $\pm$ 46	563 $\pm$ 45
120 s	529 $\pm$ 44	494 $\pm$ 30	533 $\pm$ 43	501 $\pm$ 30	539 $\pm$ 44	528 $\pm$ 42
150 s	508 $\pm$ 44	474 $\pm$ 29	514 $\pm$ 43	479 $\pm$ 29	518 $\pm$ 43	512 $\pm$ 42
180 s	500 $\pm$ 42	465 $\pm$ 29	506 $\pm$ 43	468 $\pm$ 29	508 $\pm$ 43	506 $\pm$ 42

Abbreviations: PO_{incr} , mean power output over the last 6 min stage of the incremental field tests; PO_{sp} , maximal sprint peak power output; PO_{3min} , record power output over 3 min; k_1 , exponential decay constant proposed by Weyand et al. (2006); k_2 , modified exponential decay constant determined with the individual best fits to the record power outputs using Equation (1); k_3 , modified exponential decay constant determined with the individual best fits to the record power outputs using Equation (2)

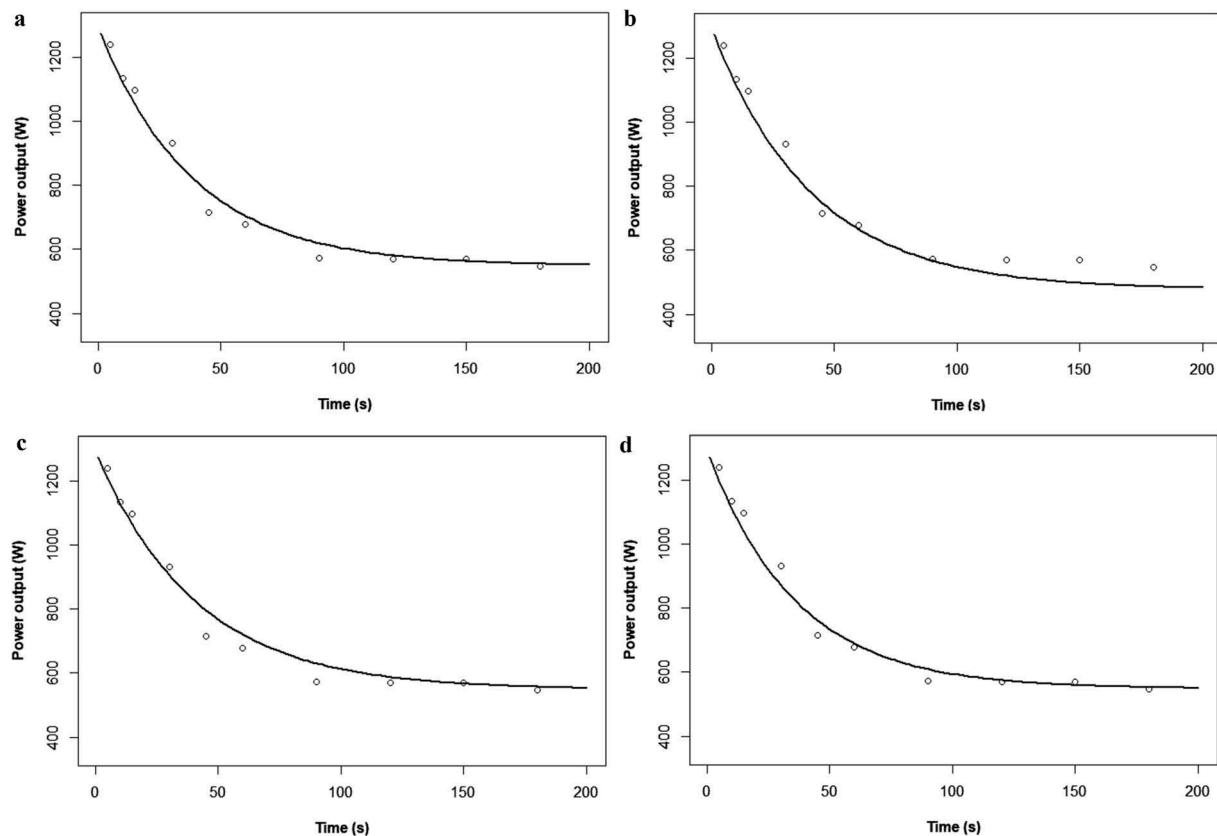


Figure 3. Power-duration curves of participant 2 showing the predicted power outputs (line) by the modelling approaches using modified approach 1 (a), modified approach 2 (b), modified approach 3 (c) and modified approach 4 (d) compared to record power outputs achieved in training and racing (open dots).

time across multiple individuals will always result in the model suiting one individual better than the other based on individual differences in power-duration decrement. Similar to the methodology used in this study, predictive ability could be improved with an exponential decay constant specific to the individual, determined using a least squares analysis. However, such an approach would only be possible if a substantial amount of (recent) data is available for that cyclist on their record power outputs over a variety of durations. It is also important to acknowledge that, as shown within the results,

the exponent may change depending on what lowest bound of the anaerobic reserve is used. If no data is available for a particular athlete, it would be advised to use a general exponential constant (e.g. $k = 0.026$) first and adjust the exponent, if needed, when a substantial amount of data on their record power outputs has been collected.

There are some limitations that need to be considered with this study. Firstly, the record power outputs achieved during the competitions are assumed to be all-out performances, it is however hard to determine if that is really the case. Race tactics and

spacing strategy as well as the influence of fatigue (e.g. during multiday stage races; Rodriguez-Marroyo, Villa, Pernia, & Foster, 2017) play an important role in the performance capability of the cyclists. However, by analysing a large number of training and competitions and determining the record power outputs they achieved over all those sessions, the influence of some of these limitations will be minimised. In addition, record power outputs were collected in the four months following the assessment of PO_{sp} and PO_{incr} . It must be acknowledged that shifts in these parameters can be expected (e.g. increase in PO_{incr} or suppression of PO_{sp} due to heavy training and/or racing) over the course of this period. However, within this current framework, irrespective of potential changes in the model parameters, the APR modelling approaches were able to predict record power outputs within a reasonable accuracy in the four months following the testing. After this period, retesting would be advised to potentially adjust model parameters and predict power output in the months following this retesting.

Practical applications

More research is needed regarding the actual practical application of this model in elite cycling, however the possible advantages that could come with the use of this model with regards to HIT prescription are promising. The lower (e.g. PO_{3min}) and upper bound (PO_{sp}) of the anaerobic reserve can be determined using two rather simple field tests. Subsequently, using the APR model, a power-duration curve from 5 to ~180 s can be established for each athlete individually. When applying HIT, a consideration should be made with regards to different work to rest ratios and the impact these may have on adaptation (Buchheit & Laursen, 2013a, 2013b). For example, a professional cyclist could be preparing for a hilly one-day race which is characterised by short hills of durations varying from 60 to 180 s and the coach would like to implement specific intervals at these durations to optimally prepare the athletes for the demands of the race. The APR model could be used to predict record power output over the interval durations, if not known for this athlete, and interval intensity could be based on a certain proportion of that record power output. Especially when the aim is to remain a steady performance over every interval (i.e. no big decreases in power output comparing the first and last interval due to fatigue) and when wanting to individualise the demand of the HIT sessions across the team, such an approach could be valuable. Especially in cycling, with multiple specialities (Lucia et al., 2001; Padilla, Mujika, Cuesta, & Goiriena, 1999) and varying competition elements (Mujika & Padilla, 2001; Padilla et al., 1999), the APR model may contribute to help coaches set the correct intensities and work to rest ratios to achieve differing competition goals.

Conclusion

To conclude, this study builds on the promising evidence of using the APR model as a tool to predict short duration (5 – 180 s) cycling performance. The decrement in all-out performance during high-intensity cycling seems to conform to a general

relationship with a single exponent appearing to be sufficient to describe the decrement in power versus duration. It is also shown that by using record power output over 3 min as a model parameter in the model, predictive ability of the model can be improved. In addition, by using an iterative least-squares method, modified exponents for a specific population of athletes can be identified to improve overall predictive ability of the model. The determination of an APR-range may contribute to the individualisation of training intensity and demand during HIT.

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